

**K. J. Somaiya College of Engineering, Mumbai-77**  
(Autonomous College Affiliated to University of Mumbai)

**End Semester Exam**  
April - May 2016

**Max. Marks: 100**

Class: SYB.Tech

Name of the Course: Object Oriented Programming Methodology

Course Code: UCEC302

**Duration: 03Hrs**

Semester: III

Branch: Computer Engg.

**Instructions:**

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

| Question No. |                                                                                                                                                                                                                                                                       | Max. Marks |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Q 1 (a)      | Explain any five JAVA features.                                                                                                                                                                                                                                       | 05         |
| Q 1 (b)      | Explain Association and Aggregation.                                                                                                                                                                                                                                  | 05         |
| Q1(c)        | Write a program to Reverser given input integer .example: input integer-123 reverser number :321                                                                                                                                                                      | 10         |
| Q2 (a)       | What are Wrapper classes in JAVA? Explain methods of wrapper classes with example.                                                                                                                                                                                    | 10         |
| Q.2 (b)      | Write a program to find out number of vowels and consonants present in the sentences also find the no of occurrences of every letter.<br>Read Multiple sentences entered by the user.<br><b>OR</b><br>Explain Methods of String and String Buffer class with example. | 10         |
| Q3 (a)       | Explain applet life cycle and write an applet program to display circle rectangle and line using applet.                                                                                                                                                              | 10         |
| Q3 (b)       | Explain two ways of creating a Thread with Example.<br><b>OR</b><br>Write a program to display ##### using multiple threads.(one thread to display # and other thread to display *)                                                                                   | 10         |
| Q4 (a)       | Define an user defined exception called "NoMatchException" that is thrown when a user input string is not equal to "Computer". Write a program with use of this exception.<br><b>OR</b><br>Explain Exception handling feature of Java with suitable example           | 10         |
| Q4 (b)       | Draw class Diagram for the Airline Reservation System.                                                                                                                                                                                                                | 10         |
| Q5 (a)       | Write a program to calculate area of rectangle, triangle and circle to demonstrate method overloading.<br><b>OR</b><br>Explain all types of inheritance with example.                                                                                                 | 10         |
| Q5(b)        | What are constructors? Explain different types of constructors with example.                                                                                                                                                                                          | 10         |